

# Euclid's Elements — Book I

## Proposition 21

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

*Comparison for segments drawn to an interior point of a triangle*

## 1 Introduction

Euclid's Proposition I.21 concerns two straight lines drawn from the ends of one side of a triangle to a point inside the triangle. Let  $ABC$  be nondegenerate. Let  $D$  be an interior point and let the line  $BD$  meet  $AC$  at  $E$ , so that

$$A - E - C, \quad B - D - E.$$

Euclid asserts simultaneously

$$BD + DC < BA + AC$$

and

$$\angle BAC < \angle BDC.$$

The present reconstruction deliberately separates these conclusions. The segment comparison is a substantial piece of synthetic segment arithmetic; the angle comparison is an independent exterior-angle argument. They are assembled only in the public theorem.

No numerical length function or numerical angle measure is introduced. Segment sums are represented by ordinary constructed segments, and strict comparisons remain the synthetic relations `HilbertSegmentLess` and `HilbertAngleLess`.

## 2 The Classical Configuration

The classical configuration uses only the original triangle and the intersection point  $E$  of the line  $BD$  with the side  $AC$ .

The single classical diagram supports two logically independent chains. For segments, Proposition I.20 is applied first to triangle  $BAE$  and then to triangle  $CDE$ . For angles, Proposition I.16 is applied to the same two auxiliary triangles.

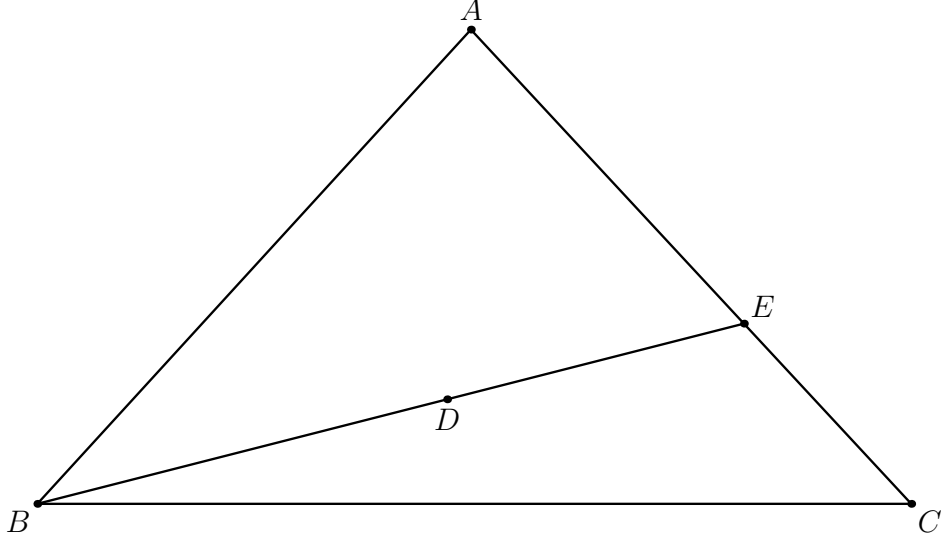


Figure 1: Classical configuration for Proposition I.21. The incidences are  $A - E - C$  and  $B - D - E$ . The same geometric state supports both the segment-sum argument and the angle-order argument.

### 3 Classical Proof

For the segment part, Proposition I.20 in triangle  $BAE$  gives

$$BE < BA + AE.$$

Adding  $EC$  geometrically gives

$$BE + EC < BA + AE + EC = BA + AC.$$

Now apply Proposition I.20 to triangle  $CDE$ . The required triangle inequality is

$$DC < DE + EC.$$

Since  $B - D - E$ ,

$$BD + DE = BE,$$

and therefore

$$BD + DC < BD + DE + EC = BE + EC < BA + AC.$$

Thus

$$BD + DC < BA + AC.$$

For the angle part, Proposition I.16 in triangle  $BAE$ , with  $AE$  extended through  $E$  to  $C$ , gives

$$\angle BAC = \angle BAE < \angle BEC.$$

Proposition I.16 in triangle  $CED$ , with  $ED$  extended through  $D$  to  $B$ , gives

$$\angle BEC = \angle CED < \angle CDB = \angle BDC.$$

Hence

$$\angle BAC < \angle BDC.$$

This audit corrects a sentence in the preserved Book1R text which stated the second triangle inequality in the form  $CE < CD + DE$ . That inequality is true, but the comparison used by the I.21 chain is the different I.20 conclusion

$$CD < CE + ED.$$

The current Lean proof uses exactly this latter form.

## 4 From Euclid's Diagram to the Formal Argument

The classical proof silently performs several kinds of operations. The formal reconstruction separates them:

- a sum such as  $BA + AC$  is represented by a concrete collinear segment rather than by a number;
- replacement of a segment by a congruent representative requires explicit transport lemmas;
- addition of unequal segments is represented by betweenness, congruence, and a theorem such as `bookZero_53_lessThanAdditive`;
- the angle proof reconstructs the exact noncollinearity and same-ray data needed by Proposition I.16.

Thus the formal proof remains synthetic, but it exposes the incidence, order, and congruence data suppressed by the classical diagram.

## 5 Synthetic Segment Arithmetic

The public theorem represents the two sums by points  $X$  and  $Y$ :

$$B - A - X, \quad AX \cong AC,$$

and

$$B - D - Y, \quad DY \cong DC.$$

Hence  $BX$  represents  $BA + AC$ , while  $BY$  represents  $BD + DC$ . The target segment assertion is the ordinary strict comparison

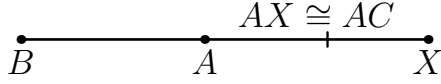
$$BY < BX.$$

This is an important structural feature of the reconstruction. There is no separate algebraic object for a segment sum. Ordinary geometric points carry the required arithmetic.

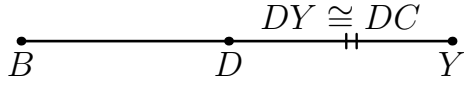
## 6 Proof Architecture of the Reconstruction

The current production file imports

```
import CGJteamLab.HilbertBookZero
import CGJteamLab.Proposition20
import CGJteamLab.Proposition16
```



$BX$  represents  $BA + AC$



$BY$  represents  $BD + DC$

Figure 2: The two public synthetic representatives in the Lean statement. The drawing records only betweenness and the copied terminal segments; the strict relation  $BY < BX$  is a theorem, not a numerical comparison read from the scale of the picture.

and contains three proposition-specific theorems:

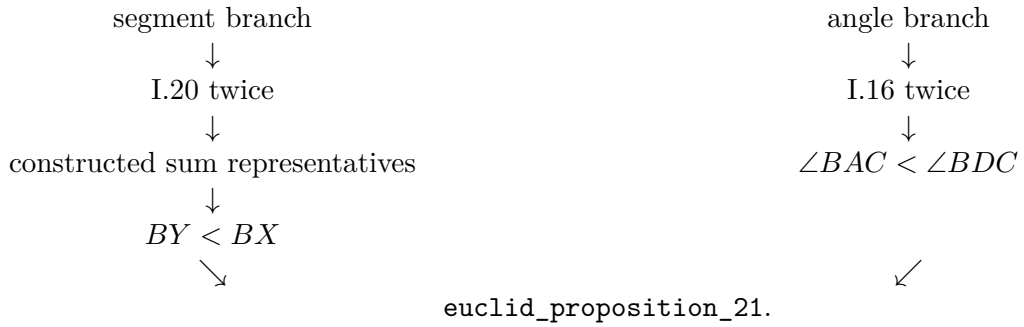
```
euclid_proposition_21_segment_helper
euclid_proposition_21_angle
euclid_proposition_21
```

All three work under

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

with no Euclidean-parallel typeclass.

The logical architecture is



The current segment branch is longer than the abbreviated description in the preserved Book1R chapter. Its actual construction chain is recorded in Figure 3.

## 6.1 The first segment-helper application

The main theorem first applies Proposition I.20 to triangle  $BAE$ . It obtains  $H$  with

$$B - A - H, \quad AH \cong AE, \quad BE < BH.$$

The theorem

```
euclid_proposition_21_segment_helper
```

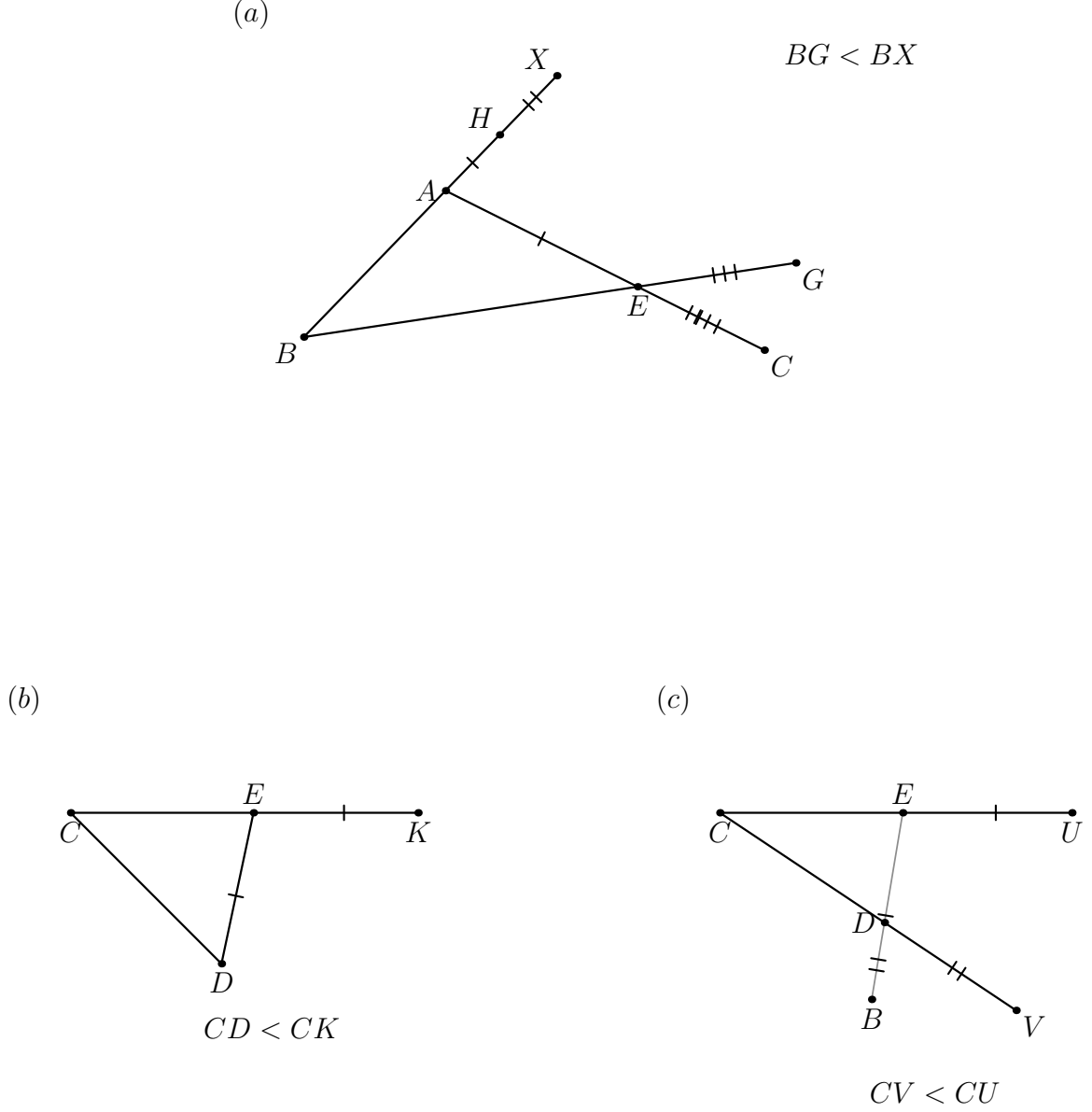


Figure 3: The proposition-specific geometric states in the current segment branch. (a) The first I.20 witness  $H$  is extended by the first segment-helper application to  $X$  and  $G$ , yielding  $BG < BX$ . (b) A second application of I.20 to triangle  $ECD$  constructs  $K$  with  $C - E - K$ ,  $EK \cong ED$ , and  $CD < CK$ . (c) The second segment-helper application constructs  $U, V$  with  $C - E - U$ ,  $C - D - V$ ,  $EU \cong EB$ ,  $DV \cong DB$ , and  $CV < CU$ . Panels (b) and (c) are deliberately separate: the proof does not need an explicit order relation between  $K$  and  $U$ .

then constructs, in the actual main theorem, points  $X$  and  $G$  such that

$$B - A - X, \quad AX \cong AC,$$

$$B - E - G, \quad EG \cong EC,$$

and

$$BG < BX.$$

Inside the generic helper the point later bound as  $X$  is named  $F$ . From  $A - E - C$  one has  $AE < AC$ . Congruence transport changes this to  $AH < AF$ , and the same-ray order theorem gives

$$A - H - F.$$

Together with  $B - A - H$  this yields the required extended collinear configuration. Difference of parts gives

$$EC \cong HF,$$

and `bookZero_53_lessThanAdditive` extends  $BE < BH$  by congruent terminal parts to the final helper conclusion.

A temporary point  $R$  is used internally only to create a suitable ray for the construction of  $G$ . It is a representation witness and does not define a new proposition-level diagram state.

## 6.2 The second I.20 state and the second helper

The public proof then applies Proposition I.20 a second time, now to triangle  $ECD$ . It obtains  $K$  with

$$C - E - K, \quad EK \cong ED, \quad CD < CK.$$

The same segment helper is then used with the roles of the points permuted. Its relevant output is a pair  $U, V$  satisfying

$$C - E - U, \quad EU \cong EB,$$

$$C - D - V, \quad DV \cong DB,$$

and

$$CV < CU.$$

The two states are geometrically distinct. Both  $K$  and  $U$  lie on the continuation determined by  $C, E$ , but the proof does not require a specific mutual order between them. They are therefore not placed on one artificially ordered diagram.

The current source contains a duplicated source-level invocation of this second helper. The first resulting pair  $U, V$  is not used; the same helper is immediately invoked again and the second pair is used in the remainder of the proof. This appears to be an incremental-proof residue rather than a distinct mathematical branch. The documentary DAG records the single logical use and no extra figure is introduced.

## 6.3 Final segment comparison

The proof recognizes the two intermediate sum representatives by congruence. In particular,

$$CU \cong BG.$$

Transport gives

$$CV < BG,$$

and transitivity with  $BG < BX$  yields

$$CV < BX.$$

Only at this later stage is the public point  $Y$  constructed with

$$B - D - Y, \quad DY \cong DC.$$

Additivity shows that  $CV$  and  $BY$  represent the same sum, hence

$$CV \cong BY.$$

A final transport through congruence gives

$$BY < BX.$$

This is exactly

$$BD + DC < BA + AC.$$

## 6.4 The angle branch

The theorem

`euclid_proposition_21_angle`

constructs no new geometric point. Its main formal work is to recover the noncollinearity data for triangles  $BAE$  and  $CED$  and then apply Proposition I.16 twice.

The exact order is

$$\angle BAC < \angle BEC = \angle CED < \angle CDB = \angle BDC.$$

The equalities in this display are implemented through same-ray and unoriented-angle transport rather than by rewriting numerical angle measures.

## 7 Lean Formalization

The public theorem is

```
theorem euclid_proposition_21
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E : Geo.Point)
  (hABC : Not (PrimCollinear Geo A B C))
  (hAEC : Geo.Between A E C)
  (hBDE : Geo.Between B D E) :
  (exists X Y : Geo.Point,
    Geo.Between B A X /\
    Geo.Congruent A X A C /\
    Geo.Between B D Y /\
    Geo.Congruent D Y D C /\
    HilbertSegmentLess Geo B Y B X) /\
  HilbertAngleLess Geo B A C B D C
```

The existential part is not auxiliary clutter. It is the exact synthetic encoding of the broken-line inequality. The second conjunct is the angle-order assertion in the established strict-angle language.

## 8 Diagrammatic Audit

The preserved Book1R chapter contains two figures, both retained here. Figure 1 is the unique classical configuration; it supports both classical proof branches and should not be duplicated merely to highlight different inequalities. Figure 2 records the public witnesses  $X, Y$  which reify the two sums.

The audit adds exactly one new figure, Figure 3, because the current Lean contains three further proposition-specific construction states which are not visible in the preserved chapter: the first I.20/helper state, the second I.20 witness  $K$ , and the second helper state  $U, V$ .

No separate diagrams are introduced for congruence symmetry, transitivity, difference of parts, inequality transport, or the angle transport steps. Those operations change proof information but do not create new incidence/order geometry. Likewise, the temporary ray witness  $R$  and the duplicated unused helper call are not diagram states.

Thus the final diagram set is

3 figures,      5 panels.

## 9 Relationship with Earlier Propositions

### 9.1 Proposition I.16

Proposition I.16 supplies the two strict exterior-angle comparisons in the angle branch. It is the principal earlier proposition behind

$$\angle BAC < \angle BDC.$$

### 9.2 Proposition I.20

Proposition I.20 is used twice in the actual segment branch. The first application to triangle  $BAE$  produces  $H$  and the inequality  $BE < BH$ . The second application to triangle  $ECD$  produces  $K$  and the inequality  $CD < CK$ . The latter application is essential to the current production DAG and should not be collapsed into a generic phrase such as “construct the remaining representative.”

### 9.3 Cumulative character

I.21 coordinates order, congruence, strict segment comparison, construction on rays, subtraction of parts, addition of inequalities, segment-comparison transitivity, I.16, and I.20. It is therefore a substantial test of Book Zero as a working synthetic proof language.

## 10 Hilbert and Book Zero Components Used

The segment branch makes direct use of, among others,

```
bookZero_30_lessThanCongruence
bookZero_32_lessThanCongruence2
bookZero_51_lessThanBetween
bookZero_3_7a
bookZero_3_7b
bookZero_differenceOfParts
bookZero_53_lessThanAdditive
bookZero_sumOfParts
bookZero_52_lessThanTransitive
```

The angle branch uses the Book Zero collinearity/noncollinearity transport layer together with the Hilbert same-ray and strict-angle transport theorems.

These results play the role that ordinary substitution and elementary segment arithmetic would play in a numerical proof, but they remain entirely synthetic.

## 11 What I.21 Reveals About Book Zero

The proof shows that an explicit algebra of lengths is not yet needed. Betweenness, congruence, segment construction, and strict comparison already permit finite sums to be represented, compared, extended, and transported.

In particular, the geometric witnesses themselves carry the arithmetic. This is exactly the intended role of Book Zero: not merely to shorten proofs, but to provide the intermediate operations in which ordinary synthetic reasoning can be expressed over the Hilbert primitives.

## 12 Dependency Summary

The current formal segment DAG is more accurately summarized as

$$\begin{array}{c}
A - E - C, B - D - E, ABC \text{ noncollinear} \\
\downarrow \\
\text{I.20 on } BAE \\
\downarrow \\
H : B - A - H, AH \cong AE, BE < BH \\
\downarrow \\
\text{first segment helper} \\
\downarrow \\
X, G : BG < BX \\
\downarrow \\
\text{I.20 on } ECD \\
\downarrow \\
K : C - E - K, EK \cong ED, CD < CK \\
\downarrow \\
\text{second segment helper} \\
\downarrow \\
U, V : CV < CU \\
\downarrow \\
\text{congruence transport and transitivity} \\
\downarrow \\
Y : B - D - Y, DY \cong DC, BY < BX.
\end{array}$$

The angle DAG is

$$\begin{array}{c}
A - E - C, B - D - E, ABC \text{ noncollinear} \\
\downarrow \\
\text{noncollinearity reconstruction} \\
\downarrow \\
\text{I.16 twice} \\
\downarrow \\
\text{same-ray and angle transport} \\
\downarrow \\
\angle BAC < \angle BDC.
\end{array}$$

The two outputs are combined only at the public endpoint `euclid_proposition_21`.

## 13 Formalization Notes

The current production source contains no local declaration named `axiom` and no `sorry`. This statement is only a source inspection result: the present documentation audit did not run `#print axioms` and therefore does not claim a fresh axiomatic audit of the endpoint.

The preserved Book1R chapter recorded that the development through I.21 compiled successfully at the time of that document. The present diagrammatic audit did not rerun `lake build`; consequently no new build claim is made here.

The main documentary correction beyond the figures is the complete second half of the segment DAG. The current Lean explicitly performs a second I.20 construction and a second segment-helper construction before building the final public witness  $Y$ .

## 14 Conclusion

Euclid I.21 is an important transition point in Book I. One classical diagram supports two different arguments, while the formal reconstruction makes their dependencies explicit.

The angle branch remains close to Euclid’s exterior-angle reasoning. The segment branch exposes a richer synthetic arithmetic of constructed segments: two uses of I.20, two logical uses of the segment helper, several distinct sum representatives, and final transport to the public witnesses  $X, Y$ .

The audited diagrams record exactly those changes of geometric state. They do not replace the synthetic proof by coordinates, numerical lengths, or numerical angle measures.